Project Report

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Course : DataBase Management System

Department : BSCS

Project Title:

Library Management System

1. Introduction

This report details the design and implementation of a **Library Management System (LMS)** database, developed as a submission for the DBMS Lab Project. The primary objective of the LMS is to modernize and automate core library operations, transitioning from error-prone manual record-keeping to an efficient, organized, and data-driven solution.

The scope of this project encompasses the management of essential library entities, including **Books**, **Authors**, and **Members**, alongside the critical function of tracking **book issuance and returns** through a comprehensive `Issue_Record`. The system is designed to uphold **data integrity** by utilizing a robust relational structure and enforcing constraints, such as unique identifiers for members' phone and email addresses. By employing industry-standard database principles, the LMS aims to provide an automated platform that enables librarians to efficiently search, update, and manage all library records

2. Objectives

The objective of the Library Management System (LMS) is to design and implement a database-driven application that enables efficient management of library operations. This includes handling book records, member registration, issuing and returning of books, maintaining fine records, and generating essential reports. The system aims to replace manual processes with an organized, automated, and error-free solution.

3. Discussion

The development of the Library Management System (LMS) database provided a successful practical demonstration of core DBMS principles. The project began with accurate conceptual modeling via the **ERD**, which was then translated into a robust, non-redundant relational schema through stringent **Normalization up to 3NF**. This process was crucial for maintaining data consistency and integrity. The final **SQL implementation** validated the design by successfully creating all tables with defined **Primary Keys** and **Foreign Keys**, enforcing **referential integrity**. The system's functionality, demonstrated by queries tracking issue and return transactions, confirms its role as an efficient, automated solution for library operations, adhering to all project requirements.

4. Scope

The system covers the following areas:

Maintaining detailed records of books, authors, and categories.

Managing member registration and membership details.

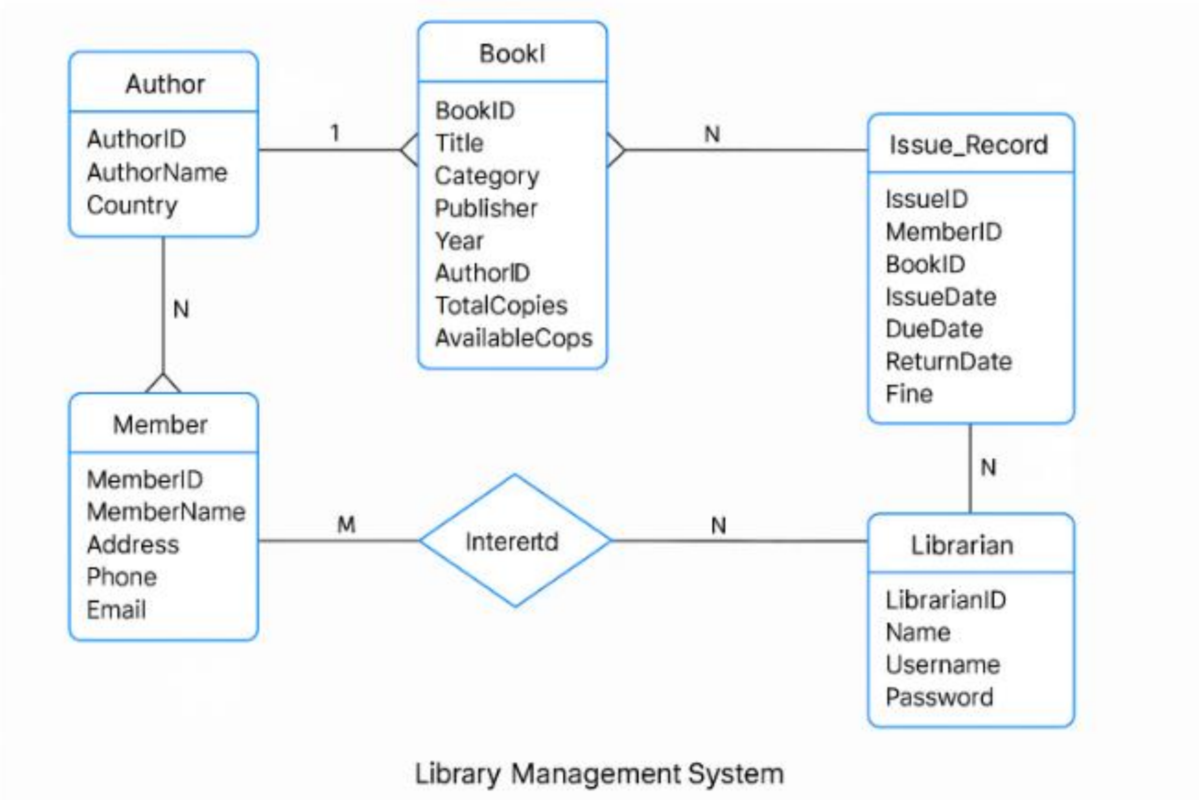
Issuing and returning of books with due date tracking.

Calculating and maintaining fine collection for overdue returns.

Allowing librarians to search, view, and update records efficiently.

Ensuring data integrity using relational database principles.

5) ERD Diagram



6. SQL Scripts

Table Creation Queries (DDL):

These scripts create the five tables and define the primary keys (PK), foreign keys (FK), and constraints. Some are :

```sql
-- 1. Create Author Table
CREATE TABLE Author (
    AuthorID INT PRIMARY KEY,
    AuthorName VARCHAR(100) NOT NULL,
    Country VARCHAR(50)
);

-- 2. Create Librarian Table
CREATE TABLE Librarian (
    LibrarianID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Username VARCHAR(50) UNIQUE,
    Password VARCHAR(50) NOT NULL
);
```

Basic Functional Queries (DML):

These queries demonstrate core database functionality.

```sql
-- Query 1: Find all available copies of books in the 'Fantasy' category.
SELECT
    Title,
    AvailableCopies
FROM
    Book
WHERE
    Category = 'Fantasy'
    AND AvailableCopies > 0;
```

7. Conclusion

1. The Library Management System project successfully applied core DBMS principles, including **ER modeling**, **normalization to 3NF**, and **SQL implementation**. This system provides a robust, consistent, and automated database solution for managing library resources, ensuring data integrity and efficient record handling. The structured approach guarantees **minimal data**

redundancy and establishes **strong referential integrity** across all relational tables.